Reprinted from THE JOURNAL OF THE ACOUSTICAL SOCIETY OF AMERICA, Vol. 31, No. 12, 1644–1648, December, 1959

Tuning Preferences for Piano Unison Groups*

ROGER E. KIRK†
The Baldwin Piano Company, Cincinnati 2, Ohio
(Received July 9, 1959)

Unison strings of a concert grand piano were tuned to five "unison" conditions. The conditions were "zero-beat" tuning and the upper string of three string unison groups tuned sharp and the lower string tuned flat by ½, 1, 2, and 3 cents relative to the center string. Magnetic tape recordings were made of the piano tuned under these conditions. These recordings in the form of a paired comparison preference test were presented to musically trained and untrained subjects. The most preferred tuning conditions for three string unison groups as recorded and reproduced from magnetic tape, are 1 and 2 cents maximum deviation among strings. Musically trained subjects prefer less deviation in tuning among unison strings than do untrained subjects. Close agreement was found between the subject's tuning preferences and the way artist tuners actually tune piano unison strings.

INTRODUCTION

THIS investigation was designed to determine the most preferred tuning condition for piano three string unison groups. Although much information has been obtained regarding the relative merits of different tuning methods and preferences for temperament,[1] little has been written concerning tuning preferences for unison groups. The dearth of literature on piano unison tuning is a logical outgrowth of assuming that piano unison groups are, as their name would imply, tuned to a *unison* condition. In the process of tuning a piano, tuners set the temperament in the octave from F_3 (175 cps) to F_4 (349 cps) and then successively tune by octaves the middle string of three string unison groups to this temperament. One of the last steps in the tuning process is the tuning of the two outer strings of a three string group to unison with the middle string. Measurements made with a Conn Chromatic Stroboscope on pianos immediately after tuning by artist tuners indicate that three string unison groups are rarely tuned to an exact unison condition. In the case of concert grand pianos, the average maximum difference among strings of a three string unison group measured at the fundamental frequency of the string is approximately 1½ cents.[2,3]

It appears from the measurements cited in the foregoing as well as from discussions with artist tuners that unison groups are deliberately not tuned to exact unison. According to piano tuners, unison groups which are tuned too closely lack interest and diminish too rapidly. A preference of musically trained subjects for some detuning of three string unison groups has been reported previously.[4] Hundley, Martin, and Benioff[5] have shown that the characteristic multiple diminution of piano tone envelope is due primarily to small differences in fundamental frequency of the unison strings. They attribute the decrease in diminution rate of the tone envelope as a function of time to two related factors: interference between strings which are almost exactly in tune, and change in rate of energy transfer from the multiple string source, before and after transition from an initial in-phase condition to a later out-of-phase condition.

It is apparent from the above considerations that some detuning of piano unison strings is desirable. The question which this research was designed to answer is how much detuning of unison strings is desirable.

PROCEDURE

In order to establish experimentally a hierarchy of preferences for unison string tuning, preliminary experiments were conducted to determine the maximum tuning difference among unison strings which would be acceptable to musically trained subjects. This difference was found to be of the order of 6–8 cents. On the basis of these preliminary experiments, five tuning conditions were selected for inclusion in a tuning preference test. The five conditions were 0, 1, 2, 4, and 6 cents difference in tuning between outside strings of unison groups. The center string of three string unison groups was tuned midway between the outside strings. It should be pointed out that in actual practice, artist tuners do not attempt to tune three string unison groups so that the center string is midway between the two outside strings. Preliminary listening tests indicated that judgments of "goodness of tuning" are related to the maximum frequency difference among the three strings rather than to the frequency spacings among the strings. The tuning procedure used in this investigation of equal frequency spacing among the strings was adopted because it provided an additional check on accuracy during the tuning process.

Tuning Method

A Baldwin grand piano style M was used in the investigation. The piano was placed in a nonreverberant

* Presented at the fifty-eighth meeting of the Acoustical Society of America, Cleveland, Ohio (October 22, 1959).
† Now at Baylor University, Waco, Texas.
[1] See R. W. Young, J. Acoust. Soc. Am. **26**, 955–959 (1954).
[2] The difference in cents between two frequencies is 1200 times $\log_2$ of their frequency ratio.
[3] Unpublished data by the writer and staff of the Acoustical Laboratory of The Baldwin Piano Company.
[4] D. W. Martin and W. D. Ward, J. Acoust. Soc. Am. **26**, 932 (1954).
[5] Hundley, Martin, and Benioff, *Proceedings of the Second International Congress on Acoustics* (American Institute of Physics, New York, 1957), paper FD4, p. 159.

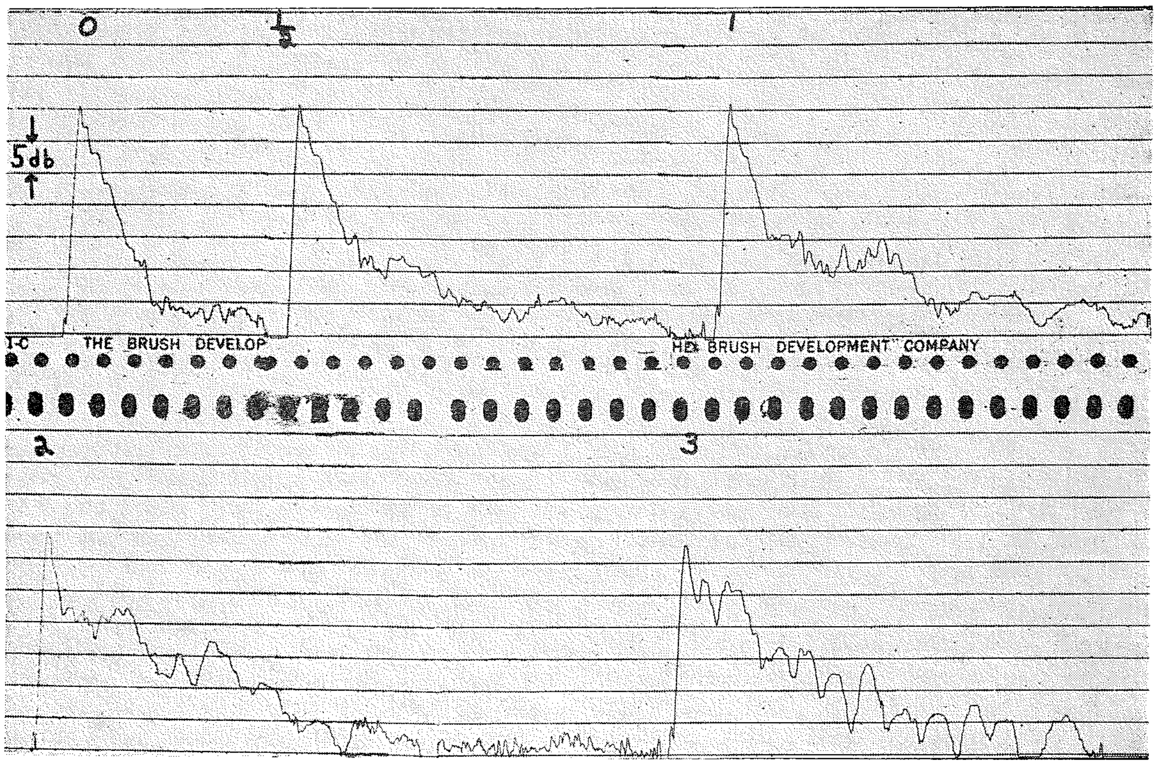

FIG. 1. Level *vs* time recordings of piano tone B_3 (247 cps). The number above the recordings refers to the tuning difference in cents between the center string and the outside strings of the unison group. These tone envelopes are representative of the envelopes produced by each of the five tuning conditions used in the preference test.

room and tuned prior to the experiment by an artist tuner. The center strings of three string unison-groups, corresponding to tones E_3 (165 cps) through G_5 (784 cps), were then checked with a Conn Chromatic Stroboscope to verify their agreement with the equally tempered scale.[6]

In order to achieve the "0 cent" tuning condition, the three string unison-groups were first tuned as close as possible to a zero-beat condition by the conventional aural method. A 640-AA Western Electric microphone was positioned 2-ft above the piano strings. Its output was amplified and led to a Brüel and Kjaer high speed level recorder. Final adjustments in the tensions of the outside strings were made while monitoring the results with the level recorder. The criterion for the zero-beat condition was the absence of amplitude fluctuations in the level recordings for two seconds after onset of the tone and for a level decrease of 30 db relative to the initial peak level of the tone.[7] Unison groups in the top $2\frac{1}{3}$ octaves of the piano were not included in this investigation. The zero-beat tuning condition there is very transitory. Attempts to achieve and maintain this condition in the upper $2\frac{1}{3}$ octaves of the piano long enough to record the tones on magnetic tape were not successful.

The 1-, 2-, 4-, and 6-cent differences between outside strings of three string unison groups were obtained with the aid of a Conn Chromatic Stroboscope. The upper string of three string unison-groups was tuned sharp and the lower string tuned flat by $\frac{1}{2}$, 1, 2, and 3 cents relative to the center string. The differences in tuning between outside strings were thus 1, 2, 4, and 6 cents, respectively.

High speed level recordings of the five "unison" conditions for the tone B_3 are shown in Fig. 1. This figure illustrates the absence of a dual diminution slope and the short duration of tones tuned to a zero-beat condition. A dual diminution slope is characteristic of piano tone.[8]

Preference Test Program Material

Three types of recorded program material were used in determining tuning preferences for piano unison groups. The three types were: individual tones E_3, B_3, F_4, C_5, and G_5, an authentic cadence chord progression in the key of F and four measures of the melody line from Schubert's Unfinished Symphony No. 8 in B Minor. The program material was recorded on magnetic

[6] In these octaves of the piano, strings are tuned approximately to the equally tempered scale. Departure from the equally tempered scale increases above and below this range according to O. H. Schuck and R. W. Young, J. Acoust. Soc. Am. **15**, 1–11 (1943).

[7] Several unison-groups could not be tuned to meet these criteria and were not included in the investigation.

[8] D. W. Martin, J. Acoust. Soc. Am. **19**, 535–541, (1947).

tape. The recording system consisted of a Western Electric 640-AA condenser microphone, Western Electroacoustic amplifier and an Ampex 300 tape recorder. The recordings were made in a nonreverberant room. The piano keys for the individual tones and melody were struck by an automatic key actuating device which was controlled from an adjoining room. The chord progressions were played by the experimenter.

The recorded tape was prepared for presentation in an A-B preference test. Each of the five tuning conditions was paired with every other tuning condition, making a total of ten paired comparison judgments for each of the five individual tones, the melody and the chord progression. Each subject thus made a total of seventy judgments, ten for each of the five individual tones, ten for melody and ten for the chord progression. The order in which the tuning conditions were presented was determined from a table of random numbers.

Subjects

One hundred and twenty-three subjects served in the experiment. One hundred and two of the subjects were students in introductory psychology classes at the University of Kentucky Northern Center. The remaining twenty-one subjects included music students from the Cincinnati College Conservatory of Music and Baldwin engineering and research personnel.

Preference Test Administration

The recorded tuning preference test was given to the subjects in groups of from seven to thirty over a period of a year and a half. All of the test administrations were made in the same room under controlled environmental conditions. The floor area of the room was 40×30 ft with a total volume of 15 600 cu ft. The acoustic treatment of the room consisted of six 36 sq ft absorptive panels suspended from the ceiling. The reverberation time of the room was approximately 1.3 sec. An Altec 800 theater loudspeaker system, an Ampex 300 tape recorder, and a Fisher 50 w power amplifier were used in the play back system for the recorded preference test. The loudspeaker system was located on a stage $1\frac{1}{2}$ ft high behind an acoustically transparent screen. The first row of subjects sat approximately 15 ft in front of the loudspeaker system. The Ampex 300 tape recorder was located in an adjoining room to minimize extraneous noise.

The level for presenting the preference test material was determined by comparison with the loudness of the same material performed with maximum effort on a grand piano. The loudness of the test material at a distance of four feet in front of the speaker system was approximately equal to the loudness observed by a performer seated in front of a grand piano playing the same test material.

The subjects were told that they were participating in a tone quality preference test. Their task was to indicate on an answer sheet which tone quality they preferred. A one-second interval of silence separated the end of the first stimulus of a pair from the onset of the second stimulus. A silent period of five seconds after termination of the second stimulus of a pair was provided during which the subjects recorded their preference.

RESULTS AND DISCUSSION

The data for the 123 subjects were pooled in determining the most preferred tuning condition for piano three string unison groups. A technique has been described by Kendall[9] for estimating the true ranking of m sets of n rankings. This technique was used to assign a rank order to the tuning conditions. The results of this analysis are shown in Table I. A statistical test described by Kendall was used to determine the significance of the rank orders in Table I. This statistic is given by

$$\left\{\left[\sum(\gamma^2)-m\sum(\gamma)+\binom{m}{2}\binom{n}{2}\right]-\frac{1}{2}\binom{n}{2}\binom{m}{2}\frac{m-3}{n-2}\right\}\frac{4}{m-2} \quad (1)$$

and is distributed as chi-square with

$$v=\frac{\binom{n}{2}m(m-1)}{(m-2)^2} \quad (2)$$

degrees of freedom. In the first formula, γ refers to the summation of scores either above or below the diagonal of an n by n table. It is evident from Table I that the most preferred tuning conditions for piano unison groups are 1 and 2 cents maximum deviation between outside strings. The 6 cent tuning condition was consistently ranked last. It will be recalled that in earlier preliminary investigations, this was the maximum tuning difference among strings which was acceptable to musically trained subjects.

TABLE I. Tuning preferences for piano three string unison groups.

Tuning condition	E_3	B_3	F_4	C_5	G_5	Chordal passage	Melodic passage
0	2[a]	4[b]	4[c]	3[c]	3[c]	4[a]	2[c]
1	1	2	3	2	1	3	1
2	3	1	1	1	2	1	3
4	4	3	2	4	4	2	4
6	5	5	5	5	5	5	5

[a] significant at 10% level.
[b] significant at 2% level.
[c] significant at 1% level.

[9] M. G. Kendall, *The Advanced Theory of Statistics* (Charles Griffin and Company, London, England, 1943), Vol. I, pp. 421–433.

It would seem on *a priori* grounds that musical training and experience would influence a subject's tuning preferences for piano three string unison groups. In order to test this hypothesis, brief musical autobiographies written by the subjects at the completion of the preference test were analyzed. The subjects were assigned to one of three categories on the basis of their musical background. The first category (A) consisted of 45 subjects who had regularly engaged in the performance of music either individually or in ensembles for a minimum of five years. The second category (B) consisted of 35 subjects with less than five years of individual or ensemble performance of music but more than one year of such experience. This category consisted primarily of subjects who had participated in high school music organizations but who had discontinued their music activities after high school. The third category (C) consisted of 27 subjects with less than one year of music performance. Sixteen of the subjects were not classified because of insufficient autobiographical information. The foregoing criteria used in categorizing the subjects are admittedly arbitrary. For purposes of this investigation, however, they appear to possess face validity although they fail to take into account the subjects' attitudes toward music or their music listening habits. Pragmatic considerations dictated the use of musical background information in classifying the subjects.

The median tuning preference of subjects in the three categories is shown in Fig. 2. There is a tendency evident in Fig. 2 for subjects whose autobiographies revealed the longest history of music study and performance to prefer the least detuning of piano three string unison groups. The statistical significance of the tuning preference differences among categories A, B, and C was evaluated by a Kruskal-Wallis one-way analysis of variance test. The differences among the categories failed to reach the 5% level of significance. The dispersion of tuning preference judgments was greatest for subjects in category C and smallest for subjects in category A. The difference in dispersion of preference judgments was significant at the 1% level. This difference in dispersion of judgments among the groups probably resulted from the operation of two factors. First, the subjects in category A may have been more interested in the recorded sounds because of their prior musical experience. This difference in interest among the three categories of subjects would manifest itself in greater alertness and ability to sustain attention for subjects in category A. The experimental design of the preference test was such that lapses in attention always resulted in greater variability in the subjects' responses. Thus, part of the difference in tuning preference variance may be error variance attributable to the experimental procedure. It seems unlikely that this is the sole explanation for the observed differences in variability. A second factor is that in many types of skills, training decreases intragroup differences. It seems likely that the smaller variability of subjects in category A is in part real and not just an artifact of the experimental design.

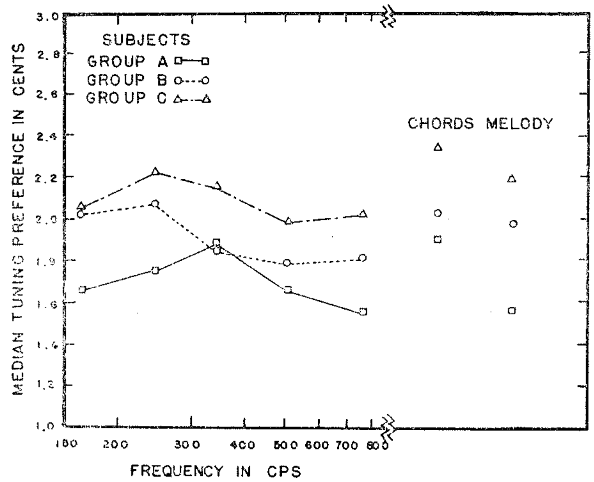

FIG. 2. Median tuning preference of subjects for individual tones, chordal passage and melodic passage. The subjects were assigned to groups A, B, and C according to their musical background. Subjects in category A had the most musical training while subjects in category C had the least musical training.

In order to determine the discriminability of the five tuning conditions, the preference test was readministered to 32 subjects in category B. The purpose of this phase of the investigation was to determine the ability of the subjects to detect a difference between the tuning conditions presented in the paired comparison test. Since a forced choice method was used in obtaining the preference judgments, it is necessary that each of the tuning conditions in comparison with the other conditions be equally discriminable. Failure to satisfy this conditions leads to ambiguity in interpretating the data. The measure of discriminability for the five tuning conditions was the subjects' ability to indicate which tone of a stimulus pair sounded smoother to them. The subjects' judgments of "smoother" were then compared with physical measurements of the tone envelopes. The percentage of judgments in which the five tuning conditions were correctly identified by the subjects as being smoother are as follows: 0 cent condition = 73%, 1 cent condition = 68%, 2 cent condition = 74%, 4 cent condition = 77%, and 6 cent condition = 87%. According to a chi-square test, the differences in correct identification associated with the five tuning conditions are not statistically significant.

It was noted earlier that piano tuners seldom tune three string unison-groups to exact unison. The mean maximum difference in fundamental frequency among strings of concert grand pianos measured immediately after tuning was 1.6 cents. The most preferred tuning condition of the subjects in category A was a mean maximum difference among unison strings of 1.7 cents. In an investigation by Corso and Lewis[10] of the pre-

[10] J. F. Corso and D. Lewis, J. Appl. Psychol. 34, 206–211 (1950).

ferred rate and extent of frequency vibrato of a complex tone, close agreement was found between the preferred extent of vibrato and the vibrato actually used by professional violinists. In their investigation, musicians preferred a narrower extent of frequency vibrato than did nonmusicians.

The results of Corso and Lewis' experiment and the present investigation raise a question concerning the role of the subjects' past experience in determining their listening preferences. We must assume that the subjects' preference judgments are determined not only by the stimulus configuration present during the test but also by the sum total of the subjects' relevant past experiences. Within this framework, the probability of evoking a listening "set" is positively correlated with the number of stimulus elements between the test situation and the subjects' previous experience which are perceived as similar at some level of awareness. It is further assumed that the level of awareness necessary for the evocation of a listening set varies from subject to subject. It has been demonstrated previously that listening to electroacoustically reproduced music with either a restricted frequency range (180–3000 cps) or an unrestricted frequency range (30–15 000 cps) results in the establishment of sets to prefer these frequency ranges.[11] This effect was greatest for musical selections similar to the selections played during the training trials and for preference tests conducted in the same environment as that used for the training trials. The importance of environmental cues in eliciting appropriate response sets has also been observed by Dulsky[12] and Abernethy.[13]

In order to determine whether the close agreement between the way pianos are tuned and listener preferences for tuning could be interpreted in terms of a set hypothesis, the preference test was readministered to 31 of the subjects in category A. The subjects were asked to indicate which tones sounded most like a piano. The mean of the distribution of judgments was 1.7 cents. It will be recalled that the mean of the distribution of preference scores for category A was also 1.7 cents. If tuning preferences are influenced by previous experience, one would predict a positive correlation between each subject's tuning preferences and his judgment of the way unison-groups are tuned in actual practice. In addition, one would predict that the correlation would be highest for subjects in category A and lowest for subjects in category C. Spearman rank-difference correlation coefficients were computed to determine the degree of relationship between tuning preferences and judgments of the way pianos are tuned. A correlation of 0.38 was obtained for 31 subjects in category A and a correlation of 0.29 for 18 subjects in category C. These correlations, while not high, are in the predicted direction and are interpreted as supporting the above set hypothesis.

[11] R. E. Kirk, J. Acoust. Soc. Am. **28**, 1113–1116 (1956).
[12] S. G. Dulsky, J. Exptl. Psychol. **18**, 725–740 (1935).
[13] E. M. Abernethy, J. Psychol. **10**, 293–301 (1940).

SUMMARY AND CONCLUSIONS

The most preferred tuning condition for piano three string unison groups was determined by means of a paired comparison preference test. Five tuning conditions were used: zero-beat tuning and the upper string of three string unison groups tuned sharp and the lower string tuned flat by $\frac{1}{2}$, 1, 2, and 3 cents relative to the center string. The preference test, recorded on magnetic tape, was given to both musically trained and untrained subjects.

An analysis of the results indicates the following:

(1) The most preferred tuning conditions for three string unison groups are 1 and 2 cents maximum deviation among strings.

(2) Subjects with the longest history of musical performance prefer the least detuning of unison groups. The intragroup variability of these subjects with respect to tuning preferences is less than the variability of subjects having little or no musical training.

(3) There is a close agreement between the average tuning preference of the subjects and the way artist tuners actually tune piano unison groups.

(4) The subjects' tuning preferences are positively correlated with the way they believe pianos sound.

(5) The findings of this investigation are interpreted as supporting a set or learning hypothesis concerning the origin of preferences for piano unison tuning.

ACKNOWLEDGMENTS

The support and stimulating suggestions of Dr. D. W. Martin, and the assistance of the writer's colleagues especially Mr. R. K. Duncan and Mr. T. C. Hundley, are gratefully acknowledged. The co-operation of Mr. T. Hankins, the faculty and student body of the University of Kentucky Northern Center is appreciated.